# The Frozen Frame

An Instrument That Measures a Language Model
Against a Ruler Made From Itself

Built by Claude instances, steered by a human, measured by machines
Toolkit: HEINRICH

The first Heinrich paper whose figures were captured, not drawn—
each rendered from the same run, in the same frame, that produced its numbers.

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**Abstract**

The previous Heinrich papers were about what models do. This one is about how to know. They shared a hidden dependency: every geometric claim was measured against a coordinate system that the measurement could itself move. A safety direction, a displacement, a trajectory—each was read off a frame re-derived from the data at hand, so the ruler flexed with the thing it measured. This paper describes the instrument built to stop that. A per-layer PCA frame is computed once over the model's own vocabulary and never re-derived; the number of retained components equals the hidden width and the components are orthonormal, so distance in the score coordinates *is* distance in the hidden state, exactly. Every token—any of the tens of thousands—then holds a fixed address, and any live forward pass can be measured against all of them at once with no truncation anywhere in the metric. The instrument declares where it cannot be trusted: a per-layer noise floor from `float16` forward reproducibility, and a frame-validity test by principal angles against a held-out projection. And it renders every measurement as a navigable surface an agent can drive and photograph, so a figure in this paper and the number beside it come from one session and one frame. We demonstrate the instrument on a single question—during a forward pass, does the residual migrate toward the token it will emit?—and answer it: no. The state does not travel to the answer; it rotates to aim at it. We close with what the instrument cannot do, and why that ceiling is a property of the method, not of the hardware.

## Contents

# 1 What This Document Is

The prior papers in this series measured behaviour: which models refuse, which break, how safety is arranged in the residual stream. HEINRICH v3 ended on a confession—every measurement error it had made pushed in the same direction, making models look safer than they were, because the investigator shared training lineage with the investigated. The lesson was not "we found the right number this time." The lesson was that a measurement instrument must be able to state its own bias, or it will quietly encode the measurer's.

This document is the instrument built from that lesson. It is not about safety, or any one model, or any one finding. It is about a ruler: how to hold a language model's internal geometry still enough to measure, how to prove the ruler has not moved, and how to publish a measurement so that its picture and its number cannot drift apart. The finding we use to exercise it—that a model *aims* at its next token rather than *travelling* to it—is a worked example, chosen because it is the cleanest thing the ruler has shown us, not because the paper is about it.

A note on the figures. This is the first Heinrich paper with any. Every earlier one was prose and tables because a hand-made figure is a claim you cannot check: the picture is drawn to illustrate a number computed elsewhere, and nothing binds them. Here an agent runs the measurement and then drives the live instrument to photograph the exact state that measurement produced. The picture and the number are the same event. Each figure carries, in monospace beneath its caption, the `.mri` it came from and the script that captured it. None of them are drawn.

# 2 The Problem: A Ruler That Moves

To compare two hidden states—this layer to that layer, this token to the answer token—you need a coordinate system they share. The usual move is to fit one from the data: PCA the activations you have, read the coordinates off the top components. The trouble is that the frame then depends on *which* activations you fed it. Add prompts, change the sample, re-run the probe, and the axes rotate; a "direction" that was there at $n$=8 is gone at $n$=15 (v3 lost its safety layers exactly this way). The ruler is a function of the thing being measured, so every distance is contaminated by the choice of measuring set. Worse for trajectories: to ask whether a live state is *near* a particular token, that token must have coordinates in the same frame—but a token you did not sample has no address at all.

What is required is a frame that (i) is fixed independently of any particular query, (ii) gives every token an address, and (iii) preserves true distances, so that "close in the picture" means "close in the model." The next section is that frame.

# 3 The Frozen Frame

Capture the model's residual state once, at every layer, over a broad token sample, and decompose each layer with PCA—but retain *all* $D$ components, where $D$ is the hidden width, and store the components $C_\ell$, the mean $\mu_\ell$, and the silence baseline $b_\ell$. This frame is written to disk and never recomputed. A live hidden state $h_\ell$ receives coordinates by the same fixed transform,

$$s \;=\; (h_\ell - b_\ell - \mu_\ell)\, C_\ell^\top, \qquad C_\ell C_\ell^\top = I, \qquad \|s - s'\|_2 \;=\; \|h_\ell - h'_\ell\|_2.$$

Because $C_\ell$ is a full-rank orthonormal basis, the projection is a lossless rotation: the score-space $L_2$ distance is the hidden-space $L_2$ distance, with no truncation approximation anywhere. The frame is not a summary of the state; it is a change of basis for it.

The vocabulary is projected through this same frozen frame—every token, not a sample—so each token id acquires a fixed per-layer address (`vocab_scores`). A live trajectory can therefore be compared, exactly, against the entire vocabulary at every depth in one pass. The frame is the coordinate contract; the `.mri` file is the artifact that carries it between the producer that captured it and any consumer that reads it.

Two subtractions are load-bearing and easy to forget. The stored mean $\mu_\ell$ recentres the frame; the silence baseline $b_\ell$ removes the model's resting state. Omitting either leaves a constant offset that, at some layers, dwarfs the signal by more than an order of magnitude. Both are applied on every projection.

## 4 The Instrument Declares Its Limits

A distance is only meaningful above the noise of the machine that produced it. HEINRICH measures that noise directly: it re-captures the sample forward pass in `float16` and compares to the stored scores, giving a per-layer reproducibility floor. Distances at or below the floor are the arithmetic of the hardware, not the model, and are drawn as such.

The frame's own validity is testable, not assumed. A second projection is derived from a held-out, crystal-suppressed vocabulary sample, and its principal angles against the frame are measured layer by layer; the variance-weighted overlap is reported as a falsification statistic. On SmolLM2-135M (raw) the frame holds at 0.90–0.999 at every layer—stated as a licence to trust the ruler, revocable the moment a model fails it.

## 5 The Shared Surface

The frame is not only a metric; it is a place. HEINRICH's companion renders the frozen frame and any live trajectory as a navigable three-dimensional surface, and it obeys one rule that makes its pictures admissible as evidence: *omit, never fake*. A segment renders only when both endpoints are genuinely in range; a candidate absent from a layer's readout is not interpolated across the gap; every view states its own sampling ("2,000 of 48,660") and its own frame ("rel to live, $s$=0.52"). The honesty is in the pixels, so a photograph of the surface carries its caveats into the PDF.

This is what lets an agent produce figures that are measurements. The agent drives the same instrument that computed the numbers, frames a view, and captures it in the same session and the same frame. Figure 1 was made exactly this way.

## 6 Worked Example: Aiming, Not Traveling

We ask the instrument one question. During a forward pass, does the final-token residual move toward the location of the token it is about to emit? The frozen frame makes this answerable exactly: the answer token has a fixed address at every layer, and so does the live state.

The proximity form of the hypothesis is false. Conditioned on the model actually producing the answer, the answer token's own frozen state sits at roughly chance rank among a background panel *even at the final layer*: the trajectory does not go there. The readout form is true. Reading each layer's state through the model's output matrix, the answer out-competes the whole panel from a late band onward and at the final layer without exception. The state does not approach the answer's location; it rotates until it projects onto the answer's output direction. Distance-to-state and preference-to-emit are different geometries—and the frozen frame is what let us hold them apart. Figure 1 is that distinction made visible: a straight spine that never bends toward a token, and a fan of readout candidates collapsing to the commit.

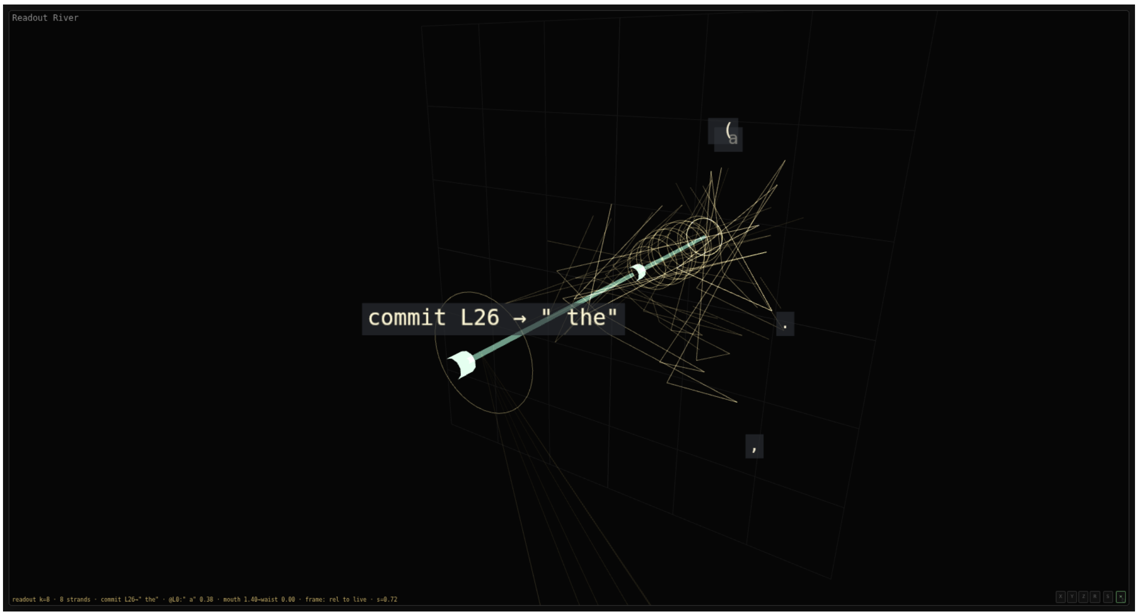


Figure 1: **The readout fan ("the horn").** SmolLM2-135M (base), prompt *"The capital of France is,"* rendered in the frozen frame. The straight teal spine is the live trajectory through depth; it runs down the axis and does not bend toward any token's location. The gold fibers are the per-layer top-$k$ readout candidates, each drawn at its true offset from the live state; they fan wide at the mouth (early layers) and collapse to the commit at the waist (`L26` $\to$ " the"). A candidate absent from the top-$k$ between two layers is not bridged—the fiber breaks rather than fake continuity. Labels resolve through the full tokenizer (whitespace shown as a visible glyph, never a raw id).

`mri: web/.data/smollm2-135m/raw.mri   capture: paper/figures/capture_horn.mjs   readout: lmhead @ h/|h|`

# 7 What the Instrument Cannot Do

The ceiling is not a matter of buying a bigger card. Faithful capture requires the residual stream at full precision at every layer; the whole methodological claim is that score distance is hidden distance *exactly*. Quantising the weights or offloading layers to fit a larger model trades that exactness away—the metric stops being lossless the moment the numbers it reads are approximate. So the honest working band is the range where full-precision internals are affordable (models up to a few billion parameters on a single 12 GB card), and the claims above are verified there and stated as unverified above it. This is a scope, not a defeat: the structural claims should hold at any scale for the same architectural reason, and the frozen frame invites replication wherever the memory exists to hold the state honestly.

A second limit is disk, not compute: the full-vocabulary projection is multiple gigabytes per model, so the machine can capture more models than it can keep captured at once. And several limits are statistical rather than physical—sample size for the scalar statistics, the construction of the background panel, the width of the noise floor—and belong in any result the instrument reports.

# 8 The Contract

# 9 Coda